

Design Token Architecture

Building a Scalable Style Foundation

The Scaling Problem

Same UI...Slightly different everywhere

No shared source of truth

No one trusts the system

What are Design Tokens?

Design Tokens are not variables

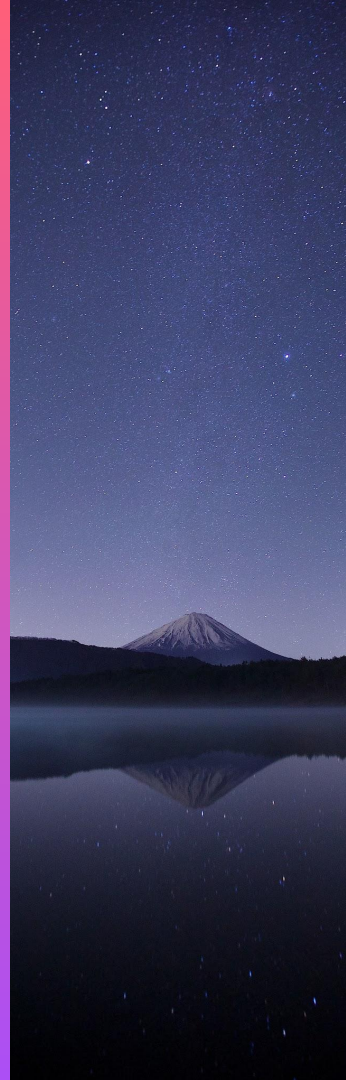
They are decisions...made reusable

The Three-Layer Token Architecture

Three-Layer Architecture

Primitive Tokens

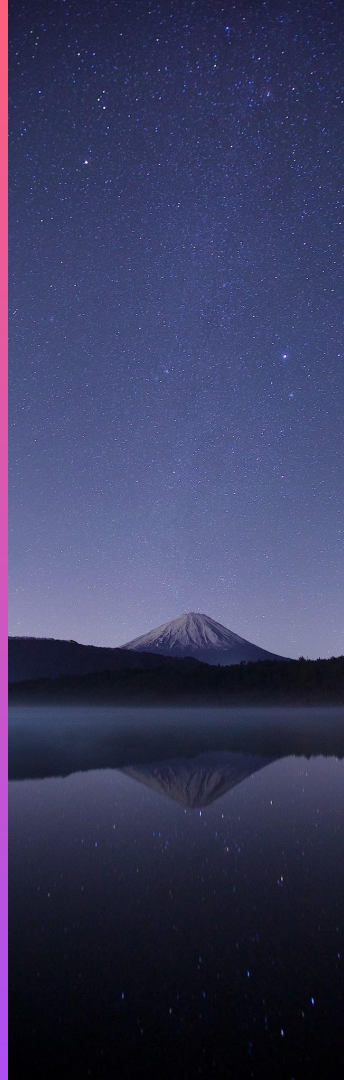
Primitives = What is



Three-Layer Architecture

Primitives

--color-blue: blue

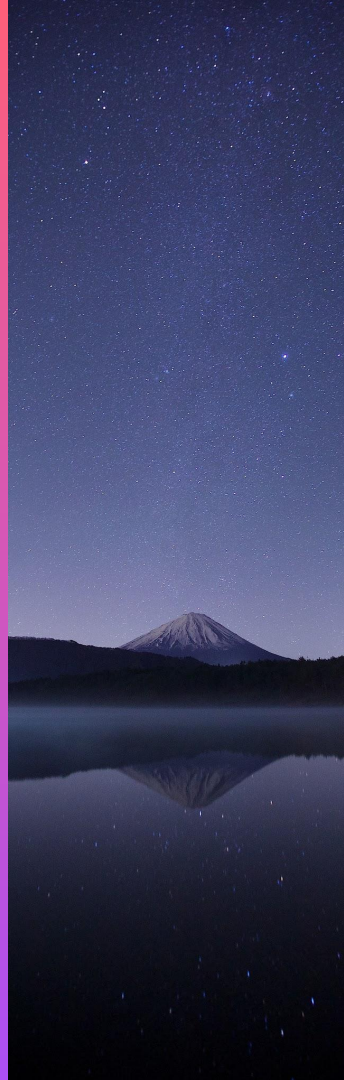


Three-Layer Architecture

Primitives

--color-blue-300: lightblue

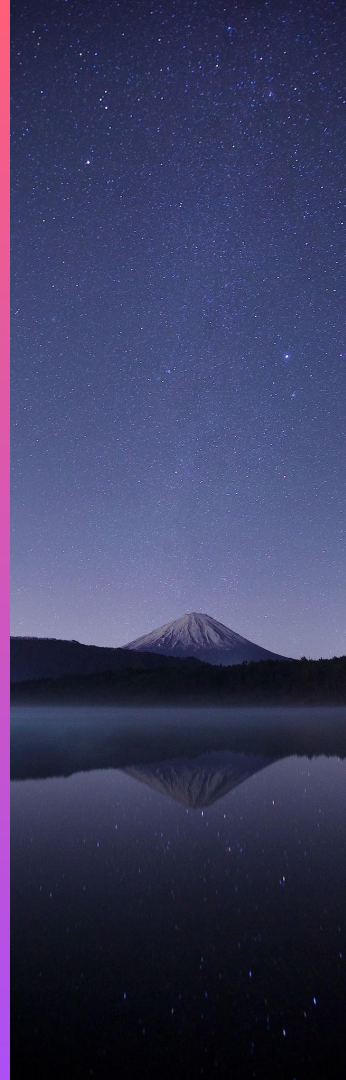
--color-blue-500: blue



Three-Layer Architecture

Semantic Tokens

Semantics = Intent



Three-Layer Architecture

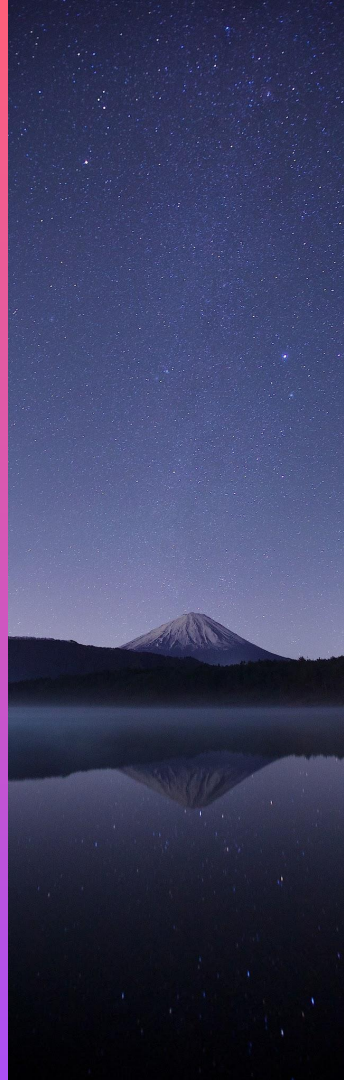
Semantics

Instead of

`--color-blue-500`

Use

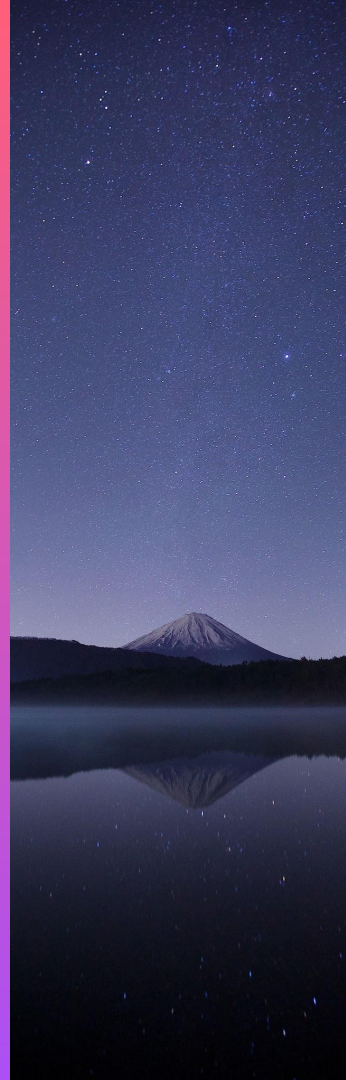
`--color-background-primary`



Three-Layer Architecture

Component Tokens

Components = Actual usage



Three-Layer Architecture

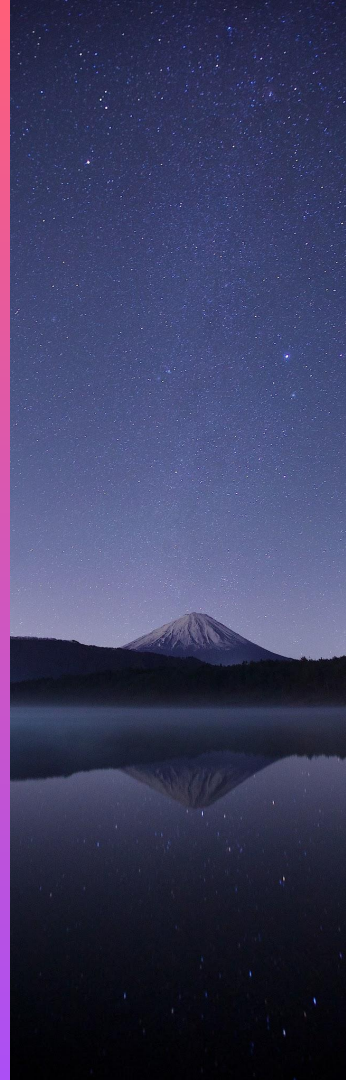
Components

Instead of

`--color-background-primary`

Use

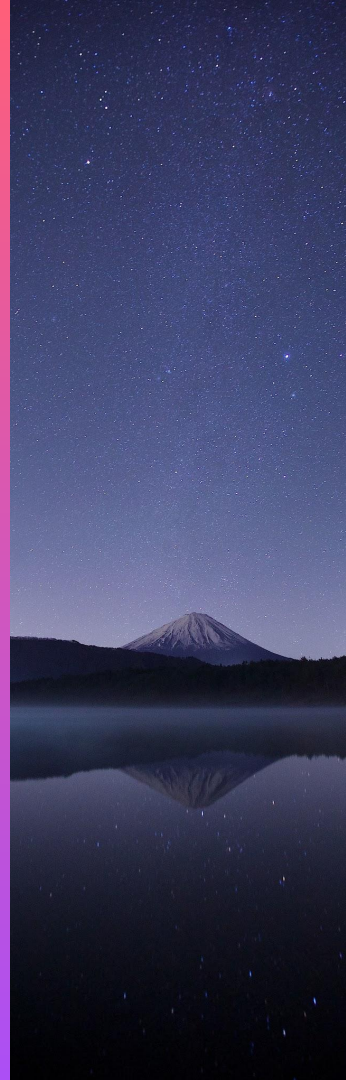
`--button-background`



Three-Layer Architecture

Components

```
background:  
var(--button-background);
```



Primitives

Semantics

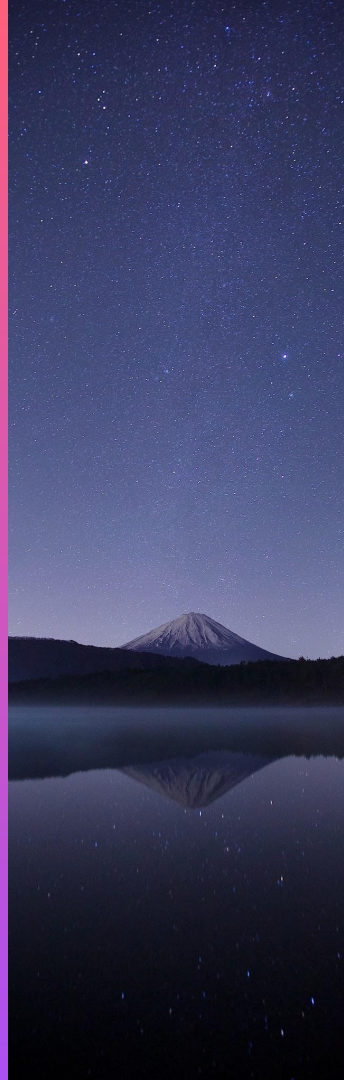
Components



What belongs in a Design System

What belongs?

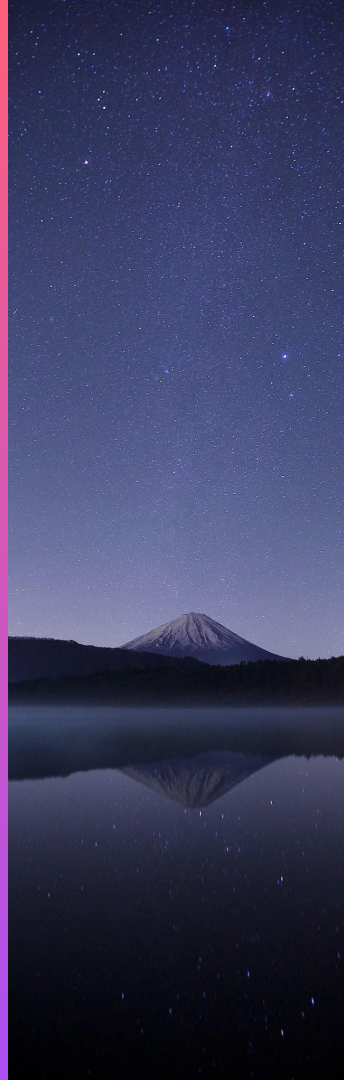
Documentation
Components
No
More Docs



The Risk of Doing Too Much/Composition Matters

The risk of doing too much

Clutter Confusion

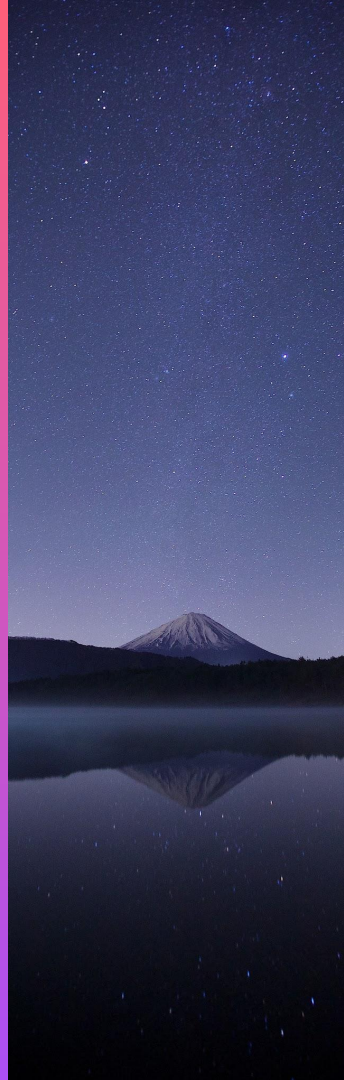




Composition is the Hallmark
of a Good System

Good Design System

Intention
Composition
Discipline

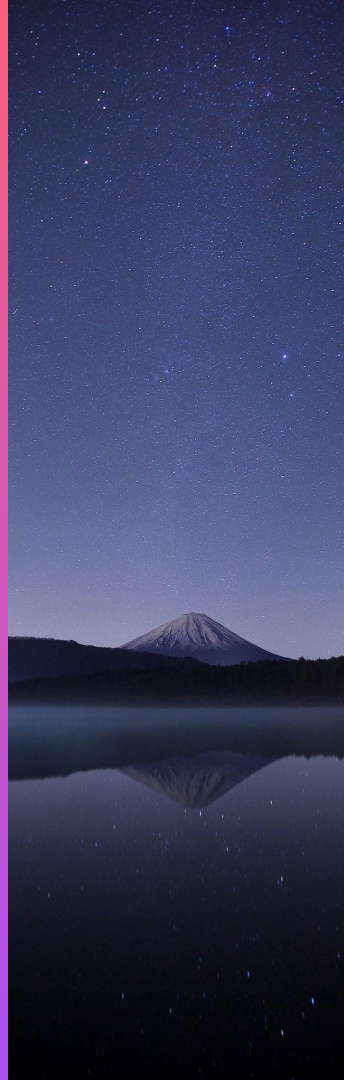


CSS Custom Properties as the Engine

CSS Custom properties

Overrides

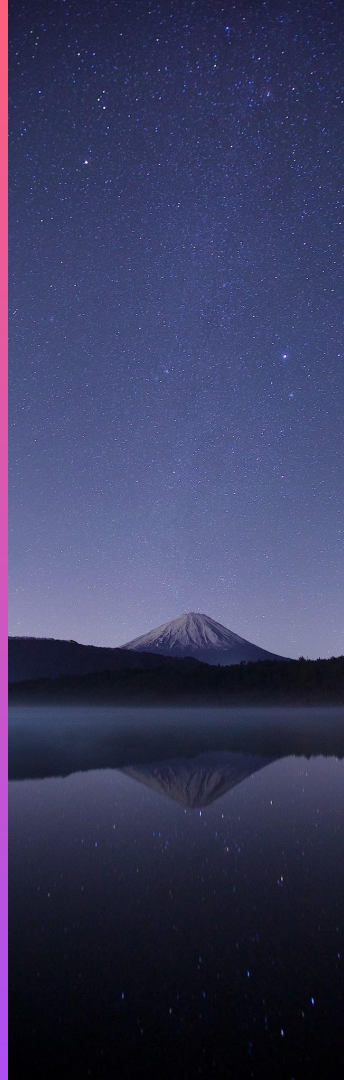
```
:root{  
  --button-background: blue;  
}  
  
.dark-mode {  
  --button-background: black;  
}
```



CSS Custom properties

Overrides

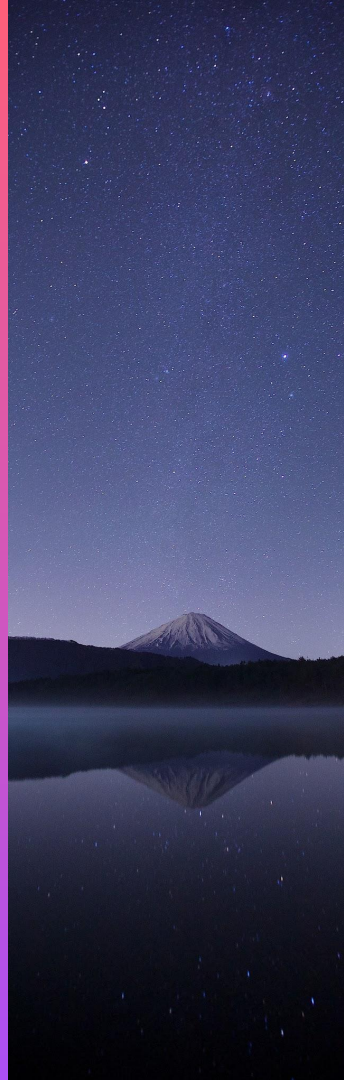
```
background:  
var(--button-background);
```



Migration Strategy

Migration Strategy

Step 1: Introduce Tokens
without deleting anything
Step 2: Map to Semantics
Step 2.5: Rinse And Repeat

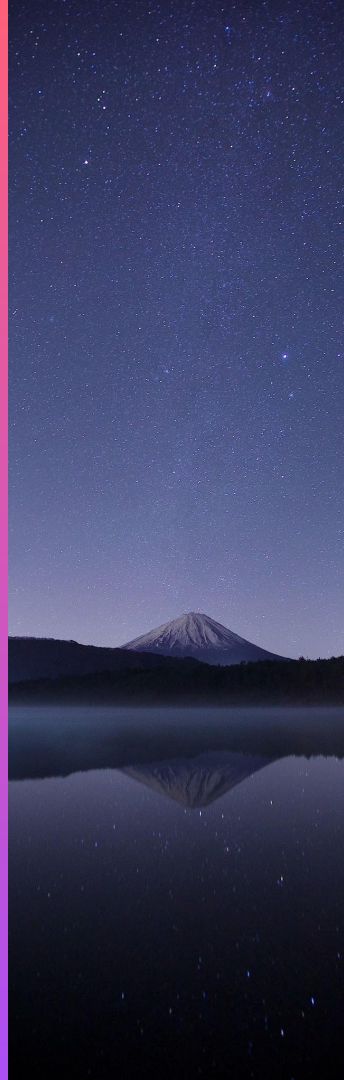


Migration Strategy

Step 3: Replace usage
gradually

```
.button {  
  height: var(--button-height);  
}
```

Step 4: Code Review/Linting



Storybook

Tokens define the language

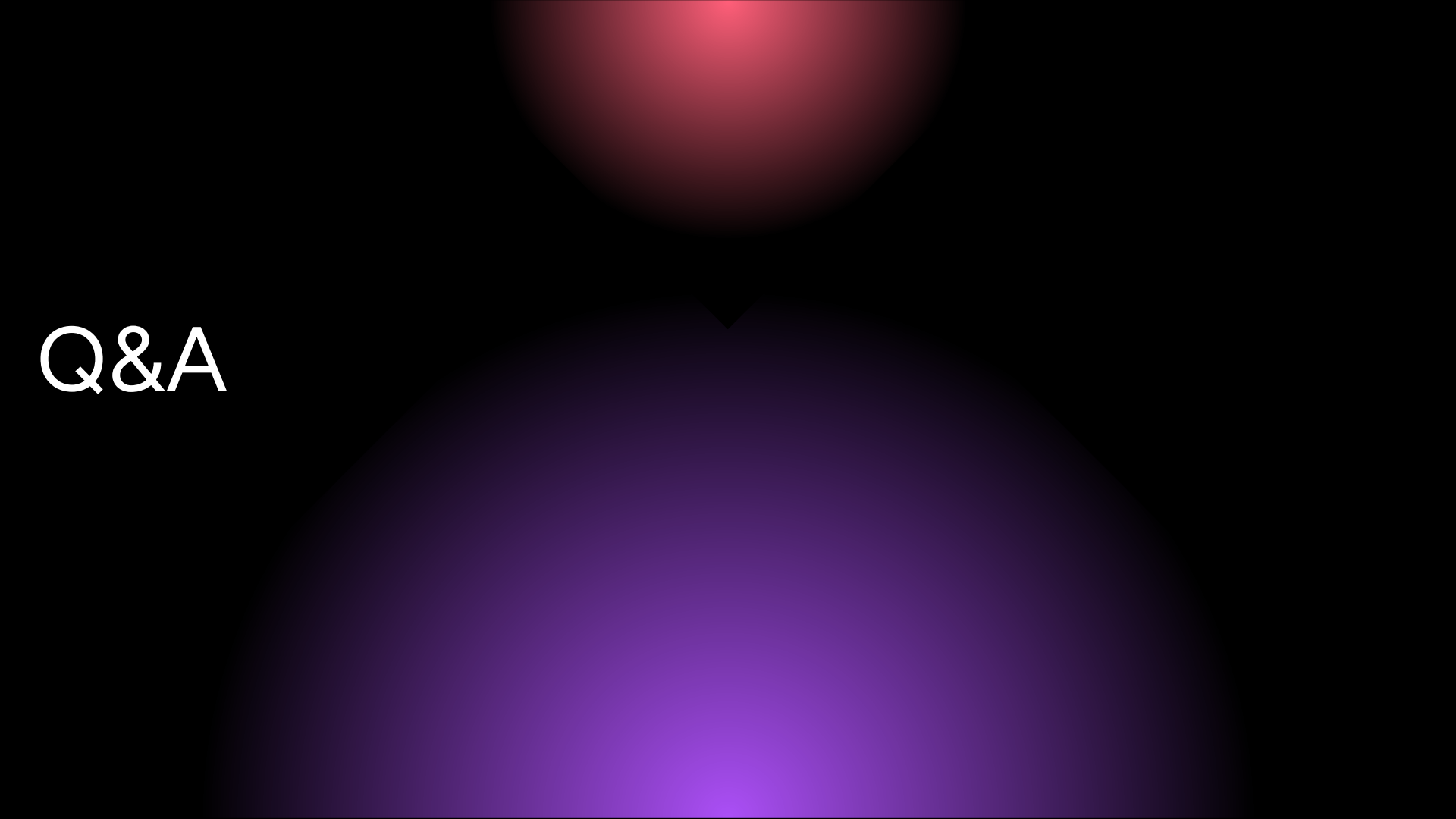
Storybook enforces it

Closing

Primitives → What is

Semantics → What it means

Components → How it's used



Q&A